

Practice Assignment

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2026-02-04

Instructions: Create a *pdf* file with all relevant code and outputs using **Rmarkdown**. Do not submit any other file formats and do not print the entire dataset with the answers. *ONLY include relevant outputs.*

Q1) Use the ‘carseats.csv’ for the data set. Read the description of the dataset and answer the following questions.

- Import the “carseats” dataset, look at the first few rows and inspect the data types of the variables in the dataframe.
- Change the variables “ShelveLoc, urban, US” into a factor variables.
- create a new variable called “profit” which stands for “Income - Advertising”
- Check for missing data. If you have missing data remove the corresponding rows from the dataset.
- How many “Good” shelving locations are there in the dataset?
- How many stores are inside the USA? create a separate data frame containing all stores from USA. Name the data set as “stores_USA”
- create another data set called “HighUrban_USSales” using ‘stores_USA’ data set. Where, sales are greater than 7 thousand and stores are located in Urban areas.
- Remove “US” and “Urban” columns from the “HighUrban_USSales” dataset.
- For one the above subset, write to a new CSV file

Q2) See the following code of a function and explain what it does. Suggest a suitable name for the function and rename. Demonstrate how the function works when you have numerical data and character data.

```
function1 <- function(x) {  
  if (length(x) <= 1) return(NULL)  
  x[-length(x)]  
}
```

Q3) Write a function to compute the sample variance of a numerical vector. Use the equation of the variance to write the function.